DENACO ANALYTICS Technical Mentor - Data Science Role Assessment

Instructions:

Please complete the following tasks to demonstrate your technical proficiency across the required tools and your ability to analyze and interpret results. Your responses should be well-structured and include explanations where necessary. Each tool will be assessed independently, so feel free to focus on the tools you are most proficient in while attempting all sections to the best of your ability. Upon completing the assessment, follow the submission guidelines below.

Submission Guidelines:

1. **Format**: Submit your assessment electronically. Use .docx, .pdf, or .txt for written responses and appropriate file formats for tool-specific outputs (e.g., .xlsx for Excel, .sql for SQL, .R for R, .py for Python, etc.).
2. **Single File**: Combine all written responses, interpretations, and screenshots into a single PDF/Word document. Include separate files for code/scripts and outputs as needed.
3. **File Naming Convention**: Name your files using the convention: "DENACO_DS_Mentor_Assessment_[YourFullName]". For example, "DENACO_DS_Mentor_Assessment_JohnDoe".
4. **Organization**: Clearly label and separate each task in your submission document as per the assessment sections.
5. **Code Execution**: For code-related tasks (R, Python, SQL, etc.), include comments explaining your logic and assumptions. Ensure your code runs without errors. Include a brief readme.txt file with instructions on how to run your code if necessary.
6. **Data Files**: Include any datasets used in a compressed folder (e.g., zip or tar.gz) named "DENACO_DS_Mentor_Dataset_[YourFullName]".
7. **Submission Deadline**: The deadline for this assessment is **Friday, 18th April 2025, 5 PM EAT**.
8. **Submission Email**: Email your completed assessment to **admin@denacoanalytics.site (mailto:admin@denacoanalytics.site)**, with a copy to **denacoent@gmail.com (mailto:denacoent@gmail.com)**. Use the subject line: "DENACO Data Science Mentor Assessment Submission - [Your Full Name]".
9. **Contact Information**: In your email, provide your full name, phone number, and email address for further communication.
10. **Academic Honesty**: Plagiarism and academic dishonesty will not be tolerated. Your work must be original, and any sources or references used should be cited appropriately.
11. **Confirmation Receipt**: You will receive a confirmation email acknowledging receipt within 24 hours. If you do not receive this, please follow up by contacting **+254757 640 352** or **+254707 661 262**.

DENACO ANALYTICS
Let Data do the Talking

Technical Assessment (Outputs & Interpretations)

Objective:

Demonstrate proficiency in the tools listed in the job description (Advanced Excel, SQL, Tableau, Power BI, SPSS, STATA, R, and Python) through technical outputs and interpretations. Each tool will be assessed independently, allowing you to showcase your strengths while attempting all sections.

Dataset Requirement:

You are required to either find suitable datasets or create your own datasets that meet the specifications provided for each section. Datasets should be relevant to a business analytics context. Ensure the dataset is included in your submission as per the guidelines. If sourcing datasets, you can use public repositories like Kaggle, UCI Machine Learning Repository, or government open data portals.

Assessment Tasks:

Section 1: Advanced Excel - Data Cleaning and Analysis

- **Dataset Specifications**: Sales data with columns: Date (YYYY-MM-DD), Region (e.g., North, South, East, West), Product (e.g., Product A, Product B, Product C), Quantity (integer), Revenue (float, in USD); ~800 rows; include ~5% missing values, ~3% duplicates, and inconsistent formats (e.g., mixed date formats).
- **Tasks**:
    1. Clean the dataset (handle missing values, remove duplicates, apply conditional formatting to highlight top 5 revenue-generating products).
    2. Create a pivot table showing revenue by region and product, and a pivot chart visualizing the results.
- **Output**: Excel file (.xlsx) with cleaned data, conditional formatting, pivot table, and chart.
- **Interpretation**: In 100-150 words, explain the pivot table/chart results and their business implications (e.g., identifying high-performing regions).

Section 2: SQL - Querying and Optimization

- **Dataset Specifications**: Two tables: Sales (SaleID, Date, Region, ProductID, Quantity, Revenue; ~800 rows) and Product Categories (ProductID, ProductName, Category e.g., Electronics, Apparel; ~20 rows); supports JOINs, subqueries, and indexing.
- **Tasks**:
    1. Write a query to calculate total revenue per region and product category, using a JOIN with the secondary table. Include a subquery to identify regions with above-average revenue.

2. Optimize the query by creating an index on the revenue column and explain the performance improvement.

- **Output**: SQL query file (.sql), screenshot of results, and screenshot of index creation.
- **Interpretation**: In 100-150 words, discuss the query results, the impact of indexing, and their relevance for sales strategies.

Section 3: Tableau - Visualization

- **Dataset Specifications**: Sales trends data with columns: Date (YYYY-MM-DD, spanning 2-3 years), Region (e.g., North, South, East, West), Product (e.g., Product A, Product B, Product C), Category (e.g., Electronics, Apparel), Revenue (float); ~1000 rows; supports time-series analysis and filtering.
- **Tasks**:

1. Create a dashboard showing revenue trends over time by region, with a filter for product categories. Include a bar chart and a line chart.

- **Output**: Tableau file (.twbx) or PDF/screenshot of the dashboard.
- **Interpretation**: In 100-150 words, describe the dashboard insights and how they could guide business decisions.

Section 4: Power BI - Data Analysis and Visualization

- **Dataset Specifications**: Product sales data with columns: Date (YYYY-MM-DD), Region (e.g., North, South, East, West), Product (e.g., Product A, Product B, Product C), Quantity (integer), Revenue (float); ~800 rows; supports slicers, DAX calculations, and KPI cards.
- **Tasks**:

1. Build a dashboard displaying quantity sold by product, with a slicer for regions. Use DAX to create a custom measure for total revenue growth percentage. Add a KPI card showing total revenue.

- **Output**: Power BI file (.pbix) or PDF/screenshot of the dashboard.
- **Interpretation**: In 100-150 words, explain the dashboard's key insights, the DAX measure, and its usefulness for a sales team.

Section 5: SPSS - Statistical Analysis

- **Dataset Specifications**: Sales comparison data with columns: Region (e.g., North, South, East, West), Product (e.g., Product A, Product B, Product C), Category (e.g., Electronics, Apparel), Revenue (float); ~500 rows; supports T-tests and chi-square tests.

- **Tasks**:
    1. Perform a T-test to compare average revenue between two regions. Conduct a chi-square test to assess the relationship between product categories and regions.
- **Output**: SPSS output file (.sav) or screenshots of results.
- **Interpretation**: In 100-150 words, interpret the statistical tests and their significance for business analysis.

Section 6: STATA - Regression Analysis

- **Dataset Specifications**: Sales regression data with columns: Product (e.g., Product A, Product B, Product C), Quantity (integer), Revenue (float); ~600 rows; supports linear regression and scatter plots.
- **Tasks**:
    1. Run a simple linear regression to predict revenue based on quantity sold. Include a scatter plot with the regression line.
- **Output**: STATA output file (.dta) or screenshots of results and scatter plot.
- **Interpretation**: In 100-150 words, discuss the regression results and their predictive value for sales forecasting.

Section 7: R - Clustering and Visualization

- **Dataset Specifications**: Regional sales data with columns: Region (e.g., North, South, East, West), Quantity (integer), Revenue (float); ~500 rows; supports k-means clustering and visualization.
- **Tasks**:
    1. Perform k-means clustering to group regions based on revenue and quantity sold. Visualize the clusters using a scatter plot.
- **Output**: R script file (.R) and visualization (e.g., scatter plot of clusters).
- **Interpretation**: In 100-150 words, explain the clustering results and their potential for targeted marketing strategies.

Section 8: Python - Machine Learning

- **Dataset Specifications**: Product performance data with columns: Product (e.g., Product A, Product B, Product C), Quantity (integer), Revenue (float), AboveAvgRevenue (binary, 1 if revenue > average, 0 otherwise); ~600 rows; supports logistic regression and visualization.
- **Tasks**:
    1. Build a logistic regression model to predict whether a product will achieve above-average revenue (binary outcome). Use pandas for data prep and matplotlib/seaborn for visualization (e.g., ROC curve).
- **Output**: Python script file (.py) and visualization.
- **Interpretation**: In 100-150 words, interpret the model's performance and its utility for product performance analysis.

Evaluation Criteria:

- **Technical Accuracy (50%)**: Correct use of tools and generation of accurate outputs.
- **Interpretation Quality (30%)**: Clarity and relevance of interpretations in a business context.
- **Completeness (20%)**: Submission of all required outputs, interpretations, and datasets for each tool.